VIT
Vellore Institute of Technology
CHENNAI

Reg. Number:	

Continuous Assessment Test (CAT) – I Jan 2026

Programme	:	B.Tech	Semester	:	Winter 2025-2026
Course Code & Course Title	:	BCSE204L & Design and Analysis of Algorithms	Class Number	:	CH2025260501125 CH2025260501138 CH2025260501145 CH2025260501149 CH2025260501151
Faculty	:	Dr. Kaja Mohideen A, Dr. Manimegalai T, Dr. Pandiyaraju V, Dr. Senthil Kumar AM, Dr. Raja Sree T	Slot	:	A1+ TA1
Duration	:	90 mins	Max. Mark		50

Answer all questions

Q. No	Sub Sec.	Description	Marks
1		You are given an unsorted array of integers. A variation of Bubble Sort, known as Odd–Even Sort (Brick Sort), operates as follows: In the odd phase, adjacent elements at index pairs (1, 2), (3, 4), (5, 6) … are compared and swapped if required. In the even phase, adjacent elements at index pairs (2, 3), (4, 5), (6, 7) … are compared and swapped if required. The algorithm alternates between odd and even phases until the array becomes sorted.	10
	a	Write the pseudocode for Odd–Even Sort indicating the odd and even phases. (4 Marks)	
	b	State whether Odd–Even Sort is a stable and comparison-based sorting algorithm. Also mention the input condition under which it achieves its best-case performance. (3 Marks)	
	c	Discuss the advantages of Odd–Even Sort over traditional Bubble Sort in a parallel processing environment. (3 Marks)	
2		A secure communication system uses a greedy-based pseudo code called HELOR for data compression before transmission. The HELOR algorithm follows a special rule during Huffman tree construction: (a) Nodes with even frequencies are assigned to the left child: (b) Nodes with odd frequencies are assigned to the right child during each merge operation. Given the following set of characters and their assumed frequencies: Character: A, E, I, D, G, O, Y, U Frequency: 8, 12, 6, 5, 9, 14, 3, 7	15
	a	Using HELOR, construct the Huffman tree for the given data set and	

		generate the Huffman code for each character. (4+2 Marks)	
	b	Encode the word "GOOD DAY" using the generated code and demonstrate the decoding process to retrieve the original message. (4 Marks)	
	c	Write a pseudo code to generate Huffman code for the characters and mention the time complexity of your pseudo code. (3+2 Marks)	
3		A company records its daily profit and loss over n days in an array A[1…n], where each element is a positive integer (profit) or negative integer (loss). Write a pseudo code to find a contiguous interval of days such that the total profit is maximized and print the profit value using the divide and conquer approach. For example, if A = [-2, 3, -1, 5, -6, 2, 4], then the profit of the contiguous interval [3, -1, 5] is 7.	15
	a	Write a pseudo code to find the maximum profit and subarray that earns the profit. (6 Marks)	
	b	Illustrate the working of the pseudo code for the given example array and calculate the profit. (7 Marks)	
	c	Determine the time complexity and justify your answer. (2 Marks)	
4		A tourism analytics company is developing a decision-support system to analyze travel packages offered across different destinations. The system performs a sequence of matrix multiplications, where each matrix represents a specific component of tourism data such as hotel availability, transport connectivity, tourist inflow, and seasonal demand. • $A_1 : 2 \times 3$ (Hotel availability data) • $A_2 : 3 \times 4$ (Transport connectivity data) • $A_3 : 4 \times 5$ (Tourist inflow data) • $A_4 : 5 \times 7$ (Seasonal demand data)	
	a	Apply the Dynamic Programming approach to compute the minimum number of scalar multiplications required to multiply the matrices – A_1, A_2, A_3, and A_4. Clearly show the intermediate calculations. (6 Marks)	
	b	Write the optimal parenthesization of the matrices – A_1, A_2, A_3, and A_4 that results in the minimum computation cost. (3 Marks)	
	c	State the time complexity of the Matrix Chain Multiplication. (1 Mark)	

**********All the best **********